

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0564

Barrett 100 kV
June 20, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0564. The Interconnection Customer proposes a 73.4 MW solar photovoltaic facility interconnecting at the Barrett 100 kV substation in the Ohio area. The request was received on January 02, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 251 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,517,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RUSSELL FORK-BLUEBONET 230kV	JONES CHAPEL-DEKALB 500kV L	154.4	900	5.5
DEKALB 230/1 kV XFMR	RICHLAND-DEKALB 500kV	143.7	1,082	6.4
STATE ST-ROBISON PARK 230kV	DEKALB-RICHLAND 500kV Line	110.7	1,617	6.8
GREENLY-DRESDEN 500kV L	INTERNATIONAL PAPER-DRESDEN 500kV	110.7	2,175	26.2
GREENLY-OMNI-HACKBERRY 500kV	INTERNATIONAL PAPER-DRESDEN 500kV	107.4	2,579	25.5
OLD FORGE-PORTAGE LAKES 230kV line	SURROWIEC-WEEDPATCH 500kV	104.6	1,030	5.6
DEKALB 500/1 kV Autotransformer	RICHLAND-DEKALB 500kV	104.6	1,488	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLUEBONNET-RUSSELL FORK 230kV	Rebuild line with higher-capacity conductor	\$68,439,000	1% (\$674,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DEKALB 230/1 kV XFMR	Replace transformer with higher-rated unit	\$12,578,000	1% (\$120,000)
STATE ST-ROBISON PARK 230kV	Rebuild line with higher-capacity conductor	\$10,533,000	1% (\$71,000)
GREENLY SUBSTATION-DRESDEN 500kV	Rebuild line with higher-capacity conductor	\$68,040,000	1% (\$803,000)
GREENLY-OMNI-HACKBERRY 500kV	Reconductor line	\$33,148,000	1% (\$396,000)
OLD FORGE-PORTAGE LAKES 230kV line	Reconductor line	\$12,071,000	1% (\$131,000)
DEKALB 500/1 kV Autotransformer	Replace transformer with higher-rated unit	\$33,632,000	1% (\$322,000)
	Total allocated to project		\$2,517,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.